

Module 02: Disk Storage and Physical Architecture

1. Physical Anatomy & Units

Platters me concentric **Tracks** hote hain jo 512B/4KB ke **Sectors** me partitioned hote hain. Sabhi platters ke identical track radius ko vertical **Cylinder** kaha jata hai.

Magnetic Hard Disk Physical Anatomy & Access Latency

Magnetic Disk Physical Structure

Spindle (Motor)

Actuator Arm

• Platter • Tracks (Concentric Rings) • Sectors

• Cylinder = Same track across all platters

Disk Access Time Formulation

1. Seek Time (T_{seek}) — LARGEST DELAY!

Read/Write head ko target track (cylinder) tak pahuchne ka samay.
Mechanical arm movement: 3 - 10 ms (Dominant factor!).

2. Rotational Latency (T_{rot})

Desired sector ko ghoomkar head ke neeche aane ka samay.
Average Rotational Latency = $1 / (2 \times \text{RPM}) \times 60 \text{ seconds}$

3. Transfer Time (T_{trans})

Data ko disk surface se controller memory me transmit karne ka time.
 $T_{trans} = (\text{Data Size}) / (\text{Transfer Rate})$

Total Disk Access Latency Formula

$T_{access} = T_{seek} + T_{rot} + T_{trans} + T_{controller}$

⇒ Goal of Disk Scheduling: Seek Time (T_{seek}) minimize karna!

2. Latency Formulation & Solved Example

Latency Component	Mathematical Formula	Example (5400 RPM, 8ms seek, 50MB/s)
Seek Time	Physical arm travel	8.00 ms (Dominant factor)
Rotational Latency	30 / RPM (seconds)	30 / 5400 = 5.56 ms
Transfer Time	Data Size / Transfer Rate	512 KB / 50 MB/s = 10.48 ms
Total Latency	Seek + Rot + Trans + Overhead	24.24 ms